Bringing Genetics Into Life: DPUK Genomics Pipeline Update

Ammar Shahabuddin[1], Alieyeh Sarabandi Moghaddam[2], Ganna Leoneko[3], Emma Squires[2], John Gallacher[1], Valentina Escott-Price[3], Sarah Bauermeister[1]

Dementias Platform UK, Department of Psychiatry, University of Oxford[1]
Dementias Platform UK, University of Swansea[2]
UK Dementia Research Institute, Cardiff University[3]

Introduction:

- **Dementias Platform UK** collects a range of biopsychosocial data from the various population cohorts
- Neurodegenerative and psychiatric disorders are complex, multifactorial conditions influenced by genetic factors.
- One approach to leverage genomic data is through the calculation of polygenic risk scores (PRS).
- However, progress in this area is currently hindered by the need to harmonise population cohorts before they can be amalgamated for analysis.

Cohort abbreviation	Cohort full name	Genetic data received	*N* SNPs	*N* samples	Phenotypes
BDR	Brains for Dementia Research	Neurochip	478,633	570	Sociodemographic, cognitive status, mental health, MMSE
GS	Generation Scotland	GS_SFHS, CHR(X), *APOE*	604,858 17,574,2	20,032 20,110	Sociodemographic, mental health, cognitive test, cognitive status
EN	EPIC Norfolk	Axiom 2020	728,244	21,041	
NSHD	Medica Research Council National Survey of Health and Development	NeuroX2 Imputed with HRC	11,081,207	2864	Amyloid status, brain measurements, MMSE
AW1 AW2	The Airwave Health Monitoring Study	Affymetrix IlluminacoreExome	845,487 542,677	4493 14,887	Sociodemographic, cognitive status, mental health

Abbreviations: *APOE*, apolipoprotein E; MMSE, Mini-Mental State Examination; SNP, single nucleotide polymorphism.

Table 1: Cohort Overview for Pipeline Creation: We implemented and tested the pipeline based on 6 of DPUKs cohorts. Joint analyses amongst cohorts provide novel insights and increases genomic density

Pipeline Creation

DPUK's initial genetic data processing pipeline includes new parallelisation techniques

- Genetic imputation (using minimac3) now conducted within 24 hours vs. 1 week of processing time

Pipeline features:

- Quality control of raw genotype data
- Imputation aligned by 1,000 Genomes reference panel
- Different methods of PRS calculation

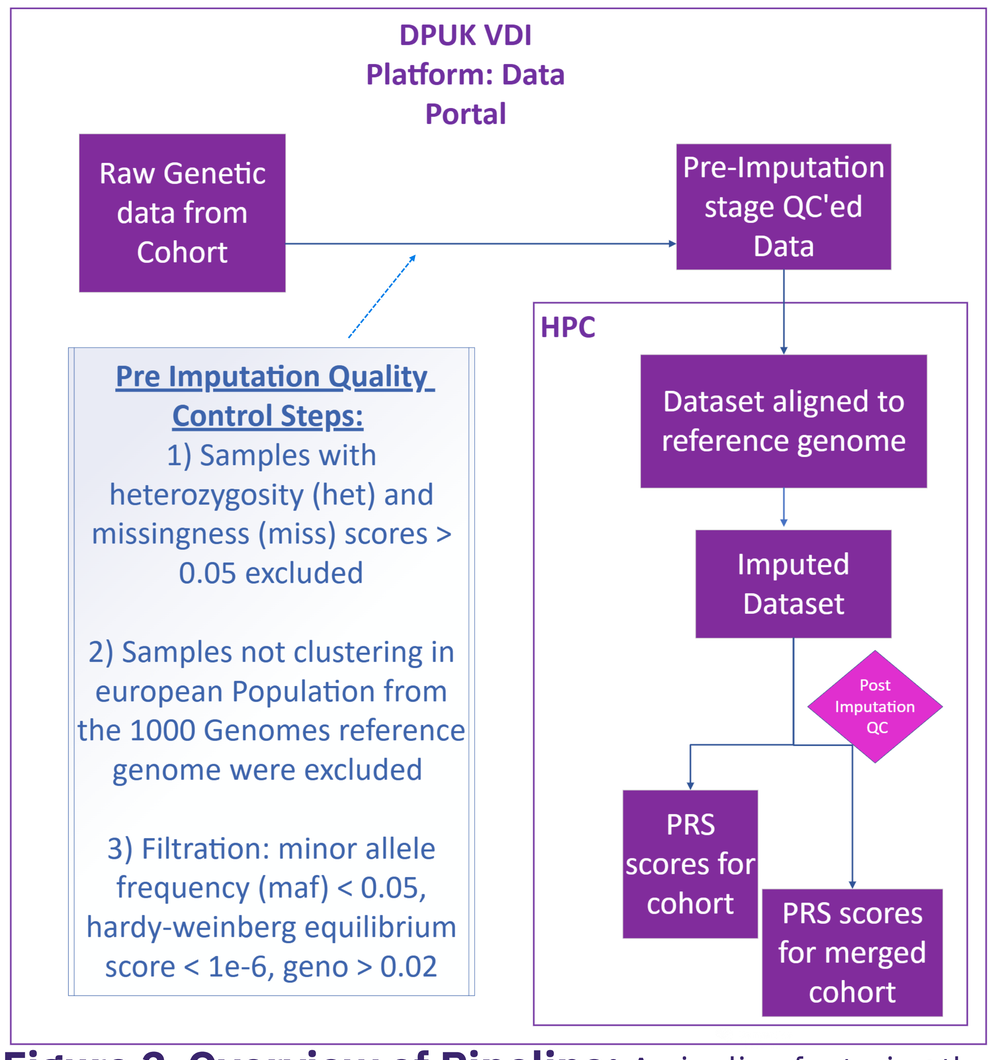

Figure 2 Overview of Pipeline: A pipeline featuring the steps one has to do within the data portal and within the high performance computer (HPC). Pipeline features faster imputation rates through parallelisation using the SLURM workload manager

Utility of the pipeline

- Providing research ready data
- Data which can be used in clinical setting
- Useful for population health prediction and management
- Ease of interpretability for non-specialists in genomics

Methods of PRS calculation

- standard clumping and threshold (PRS C+T)
- Bayesian framework (PRS-CS)

Results and Future Updates:

- Faster calculation of PRS scores for 6 neurodegenerative summary statistics for the cohorts
- Successful applicants will be provided with coding scripts that can be edited as well as documentation

Future updates

- Calculation of PRS scores for psychiatric disorders including schizophrenia and depression for cohorts with genetic data.

Further Details: ammar.shahaduddin@psych.ox.ac.uk